

Modelling the Arrival and Boarding of buses, to minimize idle waiting time.

Introduction

After dismissal from school, the primary objective of most students is to rush to the bus stop, board the bus to the train station and return home. However, due to uncertainties associated with bus arrival and boarding, students end up wasting time idly. A significant factor in the waiting time for the bus is the rate at which buses arrive. Thus, in Part 1 of my exploration I aim to model the rate of arrival of buses to the bus stop using the Poisson Distribution. In Part 2 of my exploration, I aim to model other significant factors affecting waiting time, namely the arrival of students at the bus stop from school (as seen in Figure 1) and the boarding capacity of buses. This is to find the times when students should wait for the bus or walk to the train station, and to find total student time wasted. Thus, students can make use of this information and make informed decisions of when to wait for the bus, minimising idle waiting time.

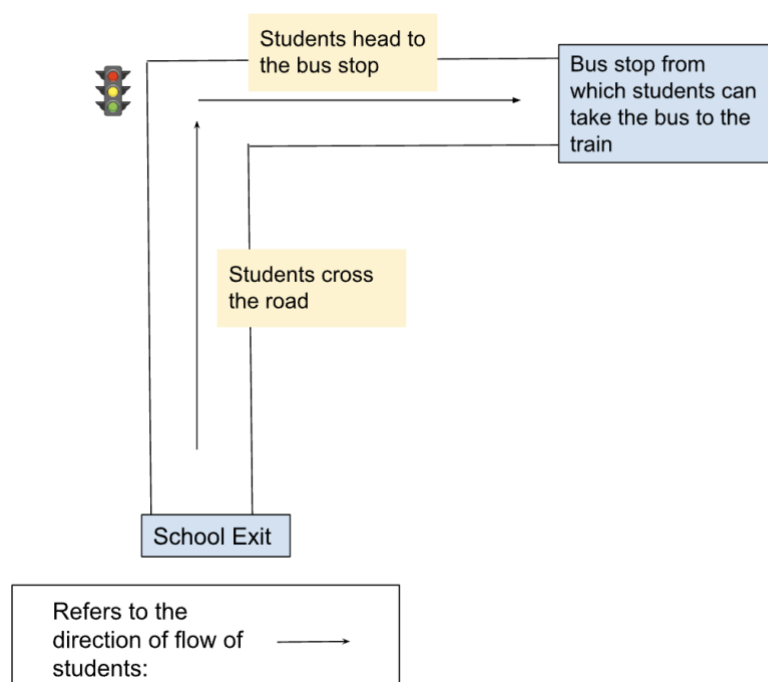


Figure 1: This is a simplified diagram of the route to reach the bus stop.

Part 1: Approximating bus arrival rate

First, using a Poisson distribution model, based upon published estimated arrival timings, I embarked upon comparing the real world data with the model to check its accuracy. A Poisson distribution is a discrete probability distribution of the probability of a number of events happening in a given time

period, given the average number of events that occurs in that time period¹. It is modelled using the below formula²,

$$P(k) = e^{-\lambda} \left(\frac{\lambda^k}{k!} \right) \text{ where, } k \in \mathbb{R}, \quad \lambda \in \mathbb{Z}^+$$

Where $P(k)$ is the probability of k number of events occurring in a fixed time interval. λ is the expected, or average number of events, in this case bus arrivals, in the fixed time interval.

The Poisson distribution makes the following assumptions³:

1. Events occur independently of each other. In this case, buses are dispatched with a constant frequency, and the arrival of one bus, does not affect the arrival of the next. Thus, it allows for the probability distribution for each event to be the same.
2. λ is constant.
3. Two events cannot occur simultaneously. In this case, no two buses can arrive at the exact same instant in time, as it is physically impossible.

According to the “land transport guru” website, the operating frequency of the two buses that service the route from the school bus stop to the train station are 5 - 14 minutes each^{4 5}. Thus, taking the average, the operating frequency of each bus is

$$\frac{5 + 14}{2} = 9.5 \approx 10 \text{ minutes per bus}$$

Thus taking a 10 minute time interval, each of the two buses would visit the bus stop once, and the expected number of bus arrivals, λ , is 2.

Thus, the probability of k buses arriving at the bus stop in a 10 minute interval is as such,

¹ Koehrsen, W. (2023, July 28). *The Poisson Process and Poisson Distribution, Explained (With Meteors!)* | Built In. Builtin.com. <https://builtin.com/data-science/poisson-process>

² Gross, J. (n.d.). *Poisson Statistics*. Retrieved October 7, 2023, from <https://web.mit.edu/jgross/Public/mit-classes/8.13/poisson-statistics-paper.pdf>

³ Koehrsen, W. (2023, July 28). *The Poisson Process and Poisson Distribution, Explained (With Meteors!)* | Built In. Builtin.com. <https://builtin.com/data-science/poisson-process>

⁴ PMPermalink, cjw19 J. 2019 at 7:49. (2017, December 31). *SBS Transit Bus Service 196 | Land Transport Guru*. Landtransportguru.net. <https://landtransportguru.net/bus196/>

⁵ *SBS Transit Bus Service 74 | Land Transport Guru*. (2017, December 31). Landtransportguru.net. <https://landtransportguru.net/bus74/>

$$P(k) = e^{-\lambda} \left(\frac{\lambda^k}{k!} \right) \text{ where, } k \in \mathbb{R}, \quad \lambda \in \mathbb{Z}^+$$

To tabulate the probabilities of bus arrival, and plot the probability distribution, Google Sheets was used as seen in figure 2.

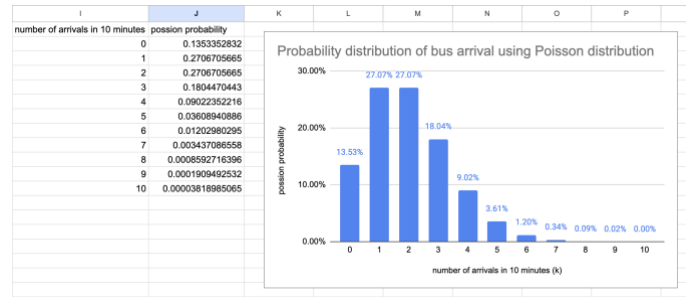


Figure 2. Use of Google Sheets to plot probability distribution

Using this aforementioned probability equation, I then plotted the following probability distribution graph in Figure 3:

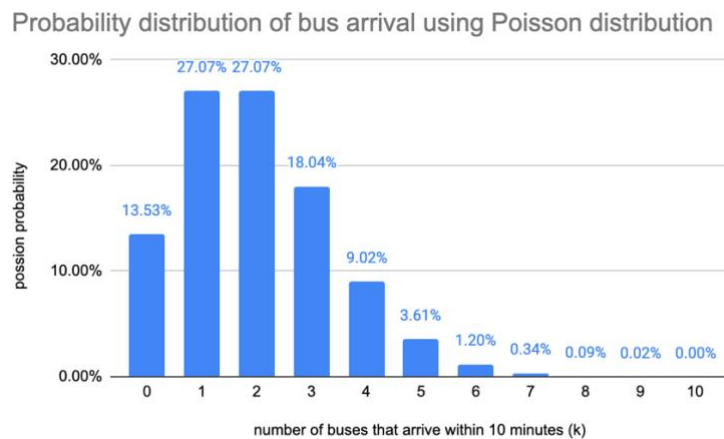


Figure 3. Probability distribution of k buses arriving in a 10 minute interval.

Since the probabilities became extremely insignificant following 10 bus arrivals, I discounted them as negligible. From the following distribution, it is evident that the probabilities of 1 or 2 buses arriving is the highest, indicating average wait times of 5 or 10 minutes for a bus.

To test the accuracy of this probability distribution, I plotted the probabilities of the bus arrivals in real time, counting the number of bus arrivals in 10 minute intervals on 6 separate days, collecting 30 data points in total, which can be seen in the appendix. Tabulating the observed number of bus arrivals in 10 minutes, against frequency, the following graph in Figure 4 was obtained.

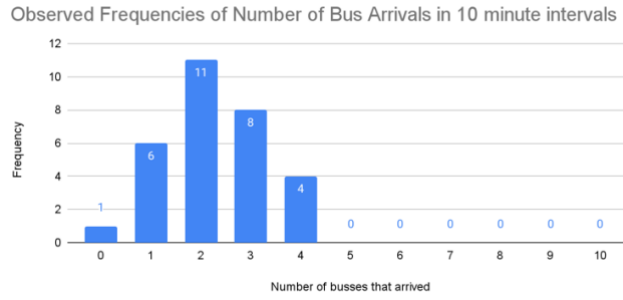


Figure 4. Observed frequency of k buses arriving in a 10 minute interval

From this distribution, we can tabulate the probabilities of k buses arriving in a 10 minute time interval by,

$$P(k) = \frac{\text{Frequency of } k \text{ buses}}{30} \times 100\%$$

For example, the probability of 2 buses arriving in a 10 minute interval is,

$$\begin{aligned} P(2) &= \frac{11}{30} \times 100\% \\ &= 36.7\% \text{ (to 3sf)} \end{aligned}$$

Plotting the probability based on observed data, the following probability distribution in Figure 5 can be obtained,

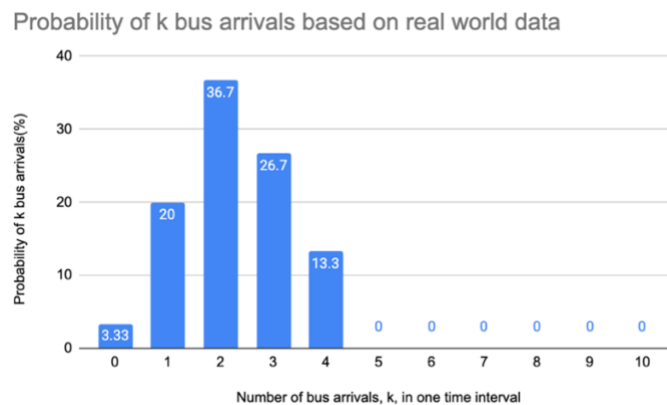


Figure 5. Probability of k bus arrivals in a 10 minute time interval based on observed data

Comparing the real world probability distribution with the Poisson distribution, it can be seen that the trend is very similar. Thus, it can be concluded that the most probable number of bus arrivals in a 10 minute interval is two. Using the finding that two buses are most probable in arriving in a 10 minute time interval, I approximated a bus arrival rate of every 5 minutes.

Part 2: Number of students at bus stop with respect to time from dismissal

Having found the most probable number of bus arrivals in a 10 minute time interval, I wanted to make this model more context specific. Especially after dismissal, the bus stop near school becomes extremely crowded and congested, with all students trying to take the bus home. Furthermore, as the bus that comes to the bus stop after dismissal is often full of students from schools down the road, students end up missing many buses, and are left waiting at the bus stop for a long time. This is a common problem faced by many of my peers, as we often do not know if it would be better for to wait for the bus to the train, or to just walk to the train station. After experiencing this multiple times, I realized that there was a pattern for when students could successfully board the bus. Thus, I modelled bus arrival and student boarding using sequences to find the optimal times for students to wait to board the bus, or to just walk to the train station, with the goal of minimising idle waiting time.

To model the pattern of students boarding the bus, the following variables and conditions were taken:

1. The time from dismissal is T where, $T = 5z, z \in \mathbb{Z}^+, 0 \leq T \leq 60$
2. The frequency the traffic light turns green is every 2.5 minutes
3. Number of students waiting at traffic light decreases with time non linearly according to the function $f(T)$
4. The number of students that can board the bus increases with time non linearly according to the function $g(T)$
5. The rate of bus arrival is every 5 minutes as determined earlier to ensure continuity in the context of this exploration
6. All students waiting at the traffic light are going to the bus stop

The conditions that I will be using for this model are as such:

1. If a student is going to be unable to board two consecutive buses, that student should walk to the train station instead. This is because walking to the station is faster only when students miss two or more buses consecutively.
2. Idle waiting refers to the time wasted when students are unable to board buses due to the crowd

To find the number of students waiting at the traffic light with respect to time from dismissal, the following data collection was done as seen in Table 1,

Time from dismissal, T (minutes)	2.5	5	7.5	10	12.5	15	17.5	20	22.5	25
Number of students	26	24	22	19	14	10	2	2	1	2

Table 1.

Plotting these values into LoggerPro data analysis software, it is visually evident that for the first 15 minutes, the values follow a quadratic pattern best, with an R^2 value of 0.9983 as seen in Figure 6.

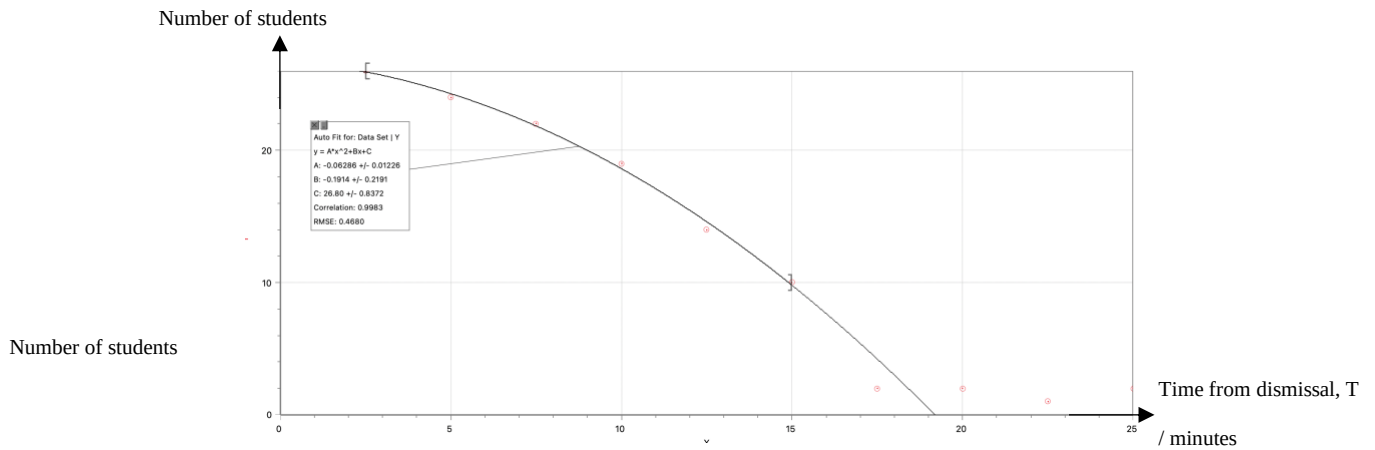


Figure 6. Quadratic best fit curve for observed data using LoggerPro software

Thus, I fitted the data into the following quadratic number pattern of students waiting at the traffic light as seen in Table 2.

Time from dismissal, T (minutes)	2.5	5	7.5	10	12.5	15	17.5	20	22.5	25
Number of students	25	24	22	19	15	10	2	2	2	2

Table 2.

From this distribution, the quadratic function, $f\left(\frac{T}{2.5}\right)$ in the form $a\left(\frac{T}{2.5}\right)^2 + b\left(\frac{T}{2.5}\right) + c$, can be found by,

$$2a = 2nd \text{ difference} = -1$$

$$a = -\frac{1}{2}$$

$$3a + b = 1st \text{ difference} = -1$$

$$b = -1 - \left(-\frac{3}{2}\right) = \frac{1}{2}$$

$$a + b + c = 1st \text{ term} = 25$$

$$c = 25 - \frac{1}{2} + \frac{1}{2} = 25$$

$$\text{Thus, } f\left(\frac{T}{2.5}\right) = -\frac{1}{2}\left(\frac{T}{2.5}\right)^2 + \frac{1}{2}\left(\frac{T}{2.5}\right) + 25, \quad \text{where } 0 \leq T \leq 15$$

For $15 < T < 25$,

$$f\left(\frac{T}{2.5}\right) = 2$$

Approximating that the number of students waiting at the bus stop remains constant at 2 until $T = 60$, the number of students **waiting at the traffic light** with respect to time from dismissal is as such,

$$f\left(\frac{T}{2.5}\right) = \begin{cases} -\frac{1}{2}\left(\frac{T}{2.5}\right)^2 + \frac{1}{2}\left(\frac{T}{2.5}\right) + 25, & 0 \leq T \leq 15 \\ 2, & 15 < T \leq 60 \end{cases}$$

Similarly, the data collected regarding the **number of students that can board the bus**, which arrives every 5 minutes, is seen in Table 3,

Time from dismissal, T (minutes)	5	10	15	20	25	30	35	40	45	50
Number of students	4	7	18	30	40	40	40	40	40	40

Table 3. Number of students that can board the bus with respect to time from dismissal

Using LoggerPro software, I found that the observed data was consistent with a quadratic number pattern with an R^2 value of 0.9945 for the first 25 minutes as seen in Figure 7.

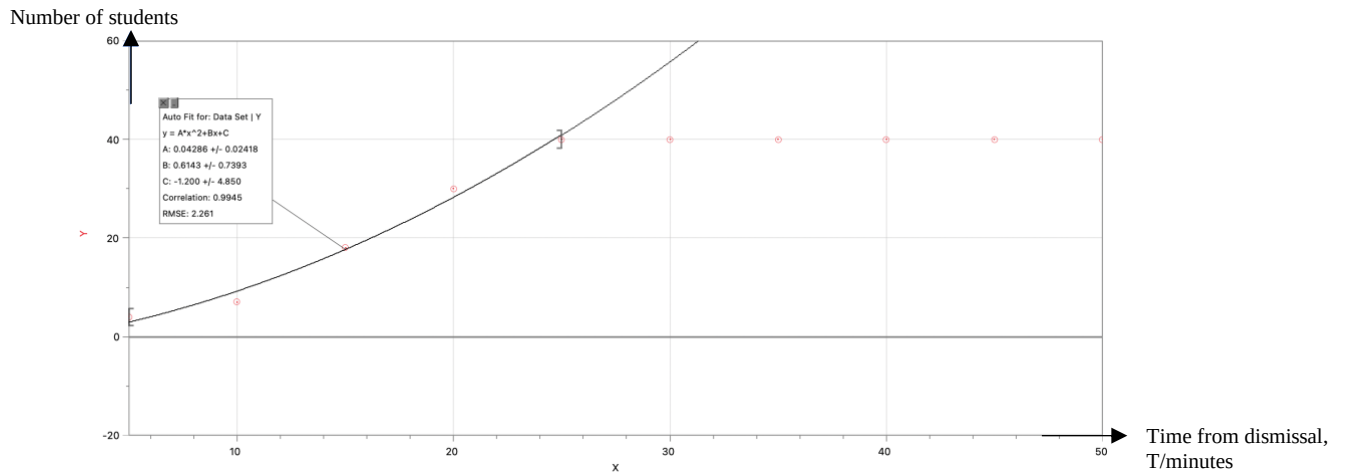


Figure 7. Quadratic best fit curve for observed data using LoggerPro software

Thus I fitted the data into a quadratic number pattern as seen in Table 4.

Time from dismissal, T (minutes)	5	10	15	20	25	30	35	40	45	50
Number of students	4	10	18	28	40	40	40	40	40	40

Table 4. Number of students that can board the bus with respect to time from dismissal

From this distribution, the quadratic function $g\left(\frac{T}{5}\right)$ in the form $a\left(\frac{T}{5}\right)^2 + b\left(\frac{T}{5}\right) + c$, can be found by,

$$2a = 2nd \text{ difference} = 2$$

$$a = 1$$

$$3a + b = 1st \text{ difference} = 6$$

$$b = 6 - 3 = 3$$

$$a + b + c = 1st \text{ term} = 4$$

$$c = 4 - 1 - 3 = 0$$

$$\text{Thus, } g\left(\frac{T}{5}\right) = \left(\frac{T}{5}\right)^2 + 3\left(\frac{T}{5}\right), \quad \text{where } 0 \leq T \leq 25$$

For $25 < T < 60$,

$$g\left(\frac{T}{5}\right) = 40$$

Thus,

$$g\left(\frac{T}{5}\right) = \begin{cases} \left(\frac{T}{5}\right)^2 + 3\left(\frac{T}{5}\right), & 0 \leq T \leq 25 \\ 40, & 25 < T \leq 60 \end{cases}$$

Having found $f(T)$ and $g(T)$ the following questions can be examined.

1. When should students wait to take the bus or walk to the train station?
2. What is the total student time wasted at the bus stop?
3. What is the probability of students being able to board the bus at a time T?

Question 1. When should students wait for the bus, or walk to the train station?

Using the sum of quadratic series⁶, the total number of students **reaching the bus stop from the traffic light** at time T, where $0 < T \leq 15$ is given by,

$$\begin{aligned}
 \sum_{r=1}^{\frac{T}{2.5}} f(r) &= \sum_{r=1}^{\frac{T}{2.5}} -\frac{1}{2}r^2 + \frac{r}{2} + 25 \\
 &= \sum_{r=1}^{\frac{T}{2.5}} -\frac{1}{2}r^2 + \sum_{r=1}^{\frac{T}{2.5}} \frac{r}{2} + \sum_{r=1}^{\frac{T}{2.5}} 25 \\
 &= -\frac{1}{2} \left[\frac{\frac{T}{2.5}}{6} \left(\frac{T}{2.5} + 1 \right) \left(2 \left(\frac{T}{2.5} \right) + 1 \right) \right] + \frac{1}{2} \left(\frac{\frac{T}{2.5}}{2} \right) \left(\frac{T}{2.5} + 1 \right) + 25 \left(\frac{T}{2.5} - 1 + 1 \right) \\
 &= -\frac{1}{2} \left[\frac{T}{15} \left(\frac{2T+5}{5} \right) \left(\frac{4T+5}{5} \right) \right] + \frac{1}{2} \left(\frac{T}{5} \right) \left(\frac{2T+5}{5} \right) + 10T \\
 &= -\frac{T}{30} \left(\frac{8T^2 + 30T + 25}{25} \right) + \left(\frac{2T^2 + 5T}{50} \right) + 10T \\
 &= \frac{-8T^3 + 7550T}{750}
 \end{aligned}$$

$\frac{T}{2.5}$ is used here as the upper limit of the series, as the traffic light turns green approximately every 2.5 minutes, as found by observation. Thus, students would reach the bus stop every 2.5 minutes. This function is only applicable for the domain of $0 < T \leq 15$, as after 15 minutes, the number of students reaching the bus stop remains relatively constant.

Another thing to note is that in real-world scenarios such as this, data is collected at discrete intervals. However, by modelling students arriving and boarding as a continuous function, it is easier to analyse and work with the data. For example, continuous functions allow the application of calculus enabling a more profound analysis of trends, which will be explored later in this exploration. Although it may be inaccurate at specific points where T is not a multiple of 5, when analysing overall trends it provides a clearer understanding of student and bus patterns.

Where $15 < T \leq 60$, the number of students reaching the bus stop from the traffic light each time the traffic light turns green is taken to be constant at around 2 students based on observations. Thus, the total number of students **reaching the bus stop from traffic light** at T, where $15 < T \leq 60$ is given by,

⁶ Easy way to find the Sum of a Quadratic Sequence using EXCEL, sigma notation | Beard Squared. (n.d.). [Www.youtube.com](https://www.youtube.com/watch?v=HV4z940tvbs). Retrieved November 11, 2023, from <https://www.youtube.com/watch?v=HV4z940tvbs>

$$\begin{aligned}
\sum_1^{\frac{T}{2.5}} f(T) &= \sum_1^{\frac{15}{2.5}} f(T) + \sum_{\frac{15}{2.5}}^{\frac{T}{2.5}} f(T) \\
&= \frac{-8(15)^3 + 7550(15)}{750} + \sum_7^{\frac{T}{2.5}} 2 \\
&= 115 + \left(\frac{T}{2.5} - 7 + 1\right)(2) \\
&= 103 + \frac{4T}{5}
\end{aligned}$$

Thus, the total number of students **arriving from the traffic light** to the bus stop, $m(T)$ is given by the piece-wise function,

$$\begin{aligned}
m(T) &= \sum_{r=1}^{\frac{T}{2.5}} f(r) \\
&= \begin{cases} \frac{-8T^3 + 7550T}{750}, & 0 < T \leq 15 \\ 103 + \frac{4T}{5}, & 15 < T \leq 60 \end{cases}
\end{aligned}$$

For example when $T = 5$, total number of students **reaching the bus stop from the traffic light** is,

$$\begin{aligned}
\sum_1^{\frac{5}{2.5}} f(T) &= \frac{-8(5)^3 + 7550(5)}{750} \\
&= 49
\end{aligned}$$

Total number of students that have **boarded the bus** at time T , where $0 \leq T \leq 25$ is given by,

$$\begin{aligned}
\sum_{r=1}^{\frac{T}{5}} g(r) &= \sum_{r=1}^{\frac{T}{5}} (r^2 + 3r) \\
&= \sum_{r=1}^{\frac{T}{5}} r^2 + \sum_{r=1}^{\frac{T}{5}} 3r \\
&= \left[\frac{\frac{T}{5}}{6} \left(\frac{T}{5} + 1 \right) \left(2 \left(\frac{T}{5} \right) + 1 \right) \right] + 3 \left(\frac{\frac{T}{5}}{2} \right) \left(\frac{T}{5} + 1 \right) \\
&= \left[\frac{T}{30} \left(\frac{T+5}{5} \right) \left(\frac{2T+5}{5} \right) \right] + 3 \left(\frac{T}{10} \right) \left(\frac{T+5}{5} \right)
\end{aligned}$$

$$\begin{aligned}
&= \frac{2T^3 + 15T^2 + 25T}{750} + \frac{3T^2 + 15T}{50} \\
&= \frac{2T^3 + 60T^2 + 250T}{750}
\end{aligned}$$

$\frac{T}{5}$ is used here as the upper limit of the series, as the bus arrives every 5 minutes, as found earlier. Thus, the number of buses that would have arrived at time T is equal to $\frac{T}{5}$. This function is only applicable for the domain of $0 \leq T \leq 25$, as after 25 minutes, the bus is relatively empty.

Since the boarding capacity of the bus each time it arrives is taken to be constant at around 40 students between 25 and 60 minutes from dismissal based on observations, **total number of students that have boarded the bus** up to time T , where $25 < T \leq 60$ is given by,

$$\begin{aligned}
\sum_1^{\frac{T}{5}} g(T) &= \sum_1^{\frac{25}{5}} g(T) + \sum_{\frac{25}{5}+1}^{\frac{T}{5}} g(T) \\
&= \frac{2(25)^3 + 60(25)^2 + 250(25)}{750} + \sum_6^{\frac{T}{5}} 40 \\
&= 100 + \left(\frac{T}{5} - 6 + 1\right)(40) \\
&= 100 + 8T - 200 \\
&= 8T - 100
\end{aligned}$$

Thus, the **total number of students boarding the bus** with respect to time from dismissal, $l(T)$, is given by the piece-wise function,

$$\begin{aligned}
l(T) &= \sum_{r=1}^{\frac{T}{5}} g(r) \\
&= \begin{cases} \frac{2T^3 + 60T^2 + 250T}{750}, & 0 < T \leq 25 \\ 8T - 100, & 25 < T \leq 60 \end{cases}
\end{aligned}$$

For example when $T = 5$, **total number of students boarding the bus** is,

$$\begin{aligned}
\sum_{r=1}^{\frac{5}{5}} g(r) &= \frac{2(5)^3 + 60(5)^2 + 250(5)}{750} \\
&= 4
\end{aligned}$$

Thus, it can be found that the general formula for **number of students remaining at the bus stop** after T minutes (assuming the bus that arrived at T minutes has left) is,

$$\sum_{r=1}^{\frac{T}{2.5}} f(r) - \sum_{r=1}^{\frac{T}{5}} g(r) = m(T) - l(T)$$

Students should walk to the train station if they would miss two buses consecutively. Thus, to find the time for which an individual student should walk to the train station rather than wait at the bus stop,

$$m(T) - l(T) > g\left(\frac{T}{5} + 1\right)$$

To compare the graphs of number of students arriving and boarding with respect to time, the functions $y = m(T) - l(T)$, and $y = g\left(\frac{T}{5} + 1\right)$ were plotted using Desmos as seen in Figure 8.

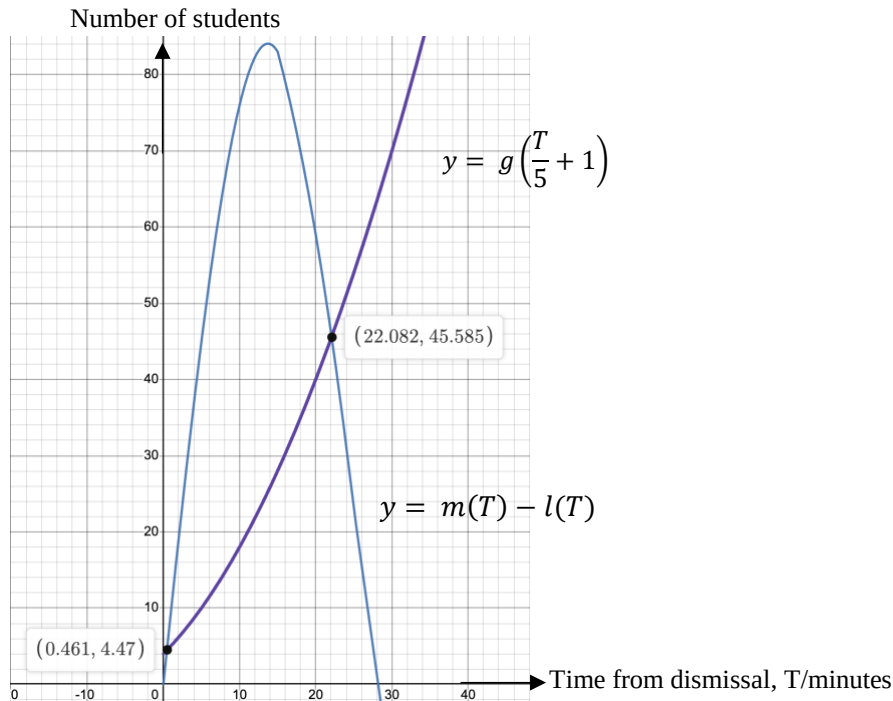


Figure 8 compares number of students at the bus stop, with capacity of the 2nd bus encountered by the student.

In Figure 8, it is evident that the graph of $y = g\left(\frac{T}{5} + 1\right)$ is greater than $y = m(T) - l(T)$ from T = 0 to T = 0.461. Since it is very difficult to leave school quickly enough to be the first few students at the bus stop, I will consider the subsequent timings. After T = 0.461, $y = g\left(\frac{T}{5} + 1\right)$ exceeds $y = m(T) - l(T)$ at T = 22.082 as seen in Figure 8. Given that the bus arrives every 5 minutes, students arriving to the bus stop after T = 17.5 should continue to wait at the bus stop, as the bus arriving at T = 25 has sufficient boarding capacity for the remaining students at the bus stop. Students arriving at T = 17.5

would miss the first bus they encounter at $T = 20$, but will be able to successfully board the 2nd bus at $T = 25$.

I had always intuitively assumed that a much shorter duration, of around 10 minutes would be required in order to catch the bus. However, when testing this model by comparing to real world experience, the results were rather accurate, as around 25 minutes from dismissal was the optimal time to board the bus. Even at 15 minutes after dismissal, I was unable to board two consecutive buses.

Question 2. What is the total student time wasted at the bus stop at a time T?

To find total student time wasted, I integrated the graph of the number of students at the bus stop against time from dismissal. But first, I needed to limit my integration to an appropriate interval, where student time is wasted only due to missing buses. Thus, I found the times at which students can board the bus, without missing any busses.

Number of students at the bus stop just before the n th bus arrives where $0 < T \leq 25$ is,

$$\sum_{r=1}^{\frac{T}{2.5}} f(r) - \sum_{r=1}^{\frac{T-5}{5}} g(r)$$

Total number of students that have boarded a bus just before the n th bus arrives where $0 < T \leq 25$,

$$\begin{aligned} \sum_{r=1}^{\frac{T-5}{5}} g(r) &= \frac{2(T-5)^3 + 60(T-5)^2 + 250(T-5)}{750} \\ &= \frac{T^3 + 15T^2 - 100T}{375} \end{aligned}$$

$\sum_{r=1}^{\frac{T-5}{5}} g(r)$ is used here as I am considering the number of students at the bus stop just before the bus arrives. Thus, an upper limit of $\frac{T-5}{5}$ is used to ensure that the students about to board are included in the total number of students waiting as well.

Total number of students that have boarded a bus just before the n th bus arrives where $25 < T \leq 60$,

$$\begin{aligned} \sum_{r=1}^{\frac{T-5}{5}} g(r) &= \sum_{r=1}^{\frac{25-5}{5}} g(r) + \sum_{r=\frac{25-5}{5}+1}^{\frac{T-5}{5}} 40 \\ &= \frac{(25)^3 + 15(25)^2 - 100(25)}{375} + \sum_{r=5}^{\frac{T-5}{5}} 40 \\ &= 60 + \left(\frac{T-5}{5} - 5 + 1 \right) (40) \\ &= 60 + 8T - 200 \\ &= 8T - 140 \end{aligned}$$

To minimize waiting time, students should leave school such that when they arrive at the bus stop, number of students at bus stop = bus capacity. Thus,

$$\sum_{r=1}^{\frac{T}{2.5}} f(r) - \sum_{r=1}^{\frac{T-5}{5}} g(r) = g\left(\frac{T}{5}\right)$$

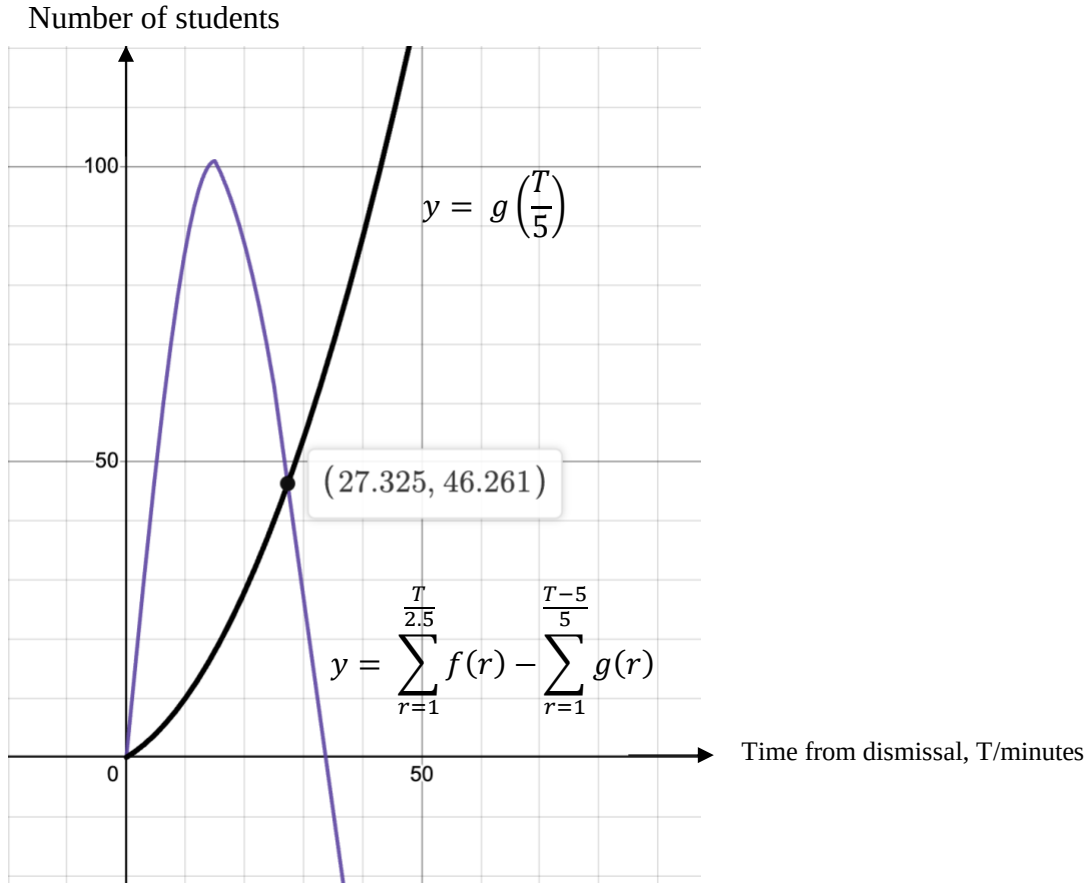


Figure 9 compares when number of students at the bus stop is equal to the capacity of the 1st bus encountered.

From here, it is evident that the time at which the number of students at the bus stop is equal to the capacity of the bus is at $T = 27.325$, since $T=0$ is impossible as students cannot possibly reach the bus stop at $T=0$. Thus, for $T > 27.325$, the capacity of the bus exceeds the number of students waiting at the bus stop, and there is no idle waiting time due to missing buses. Since T is a multiple of 5, and for $T > 27.325$, waiting time is no longer significant, I have limited my calculation of total student time wasted in the interval $0 \leq T \leq 25$, as any time past 25 minutes, students will be able to board the first bus they encounter.

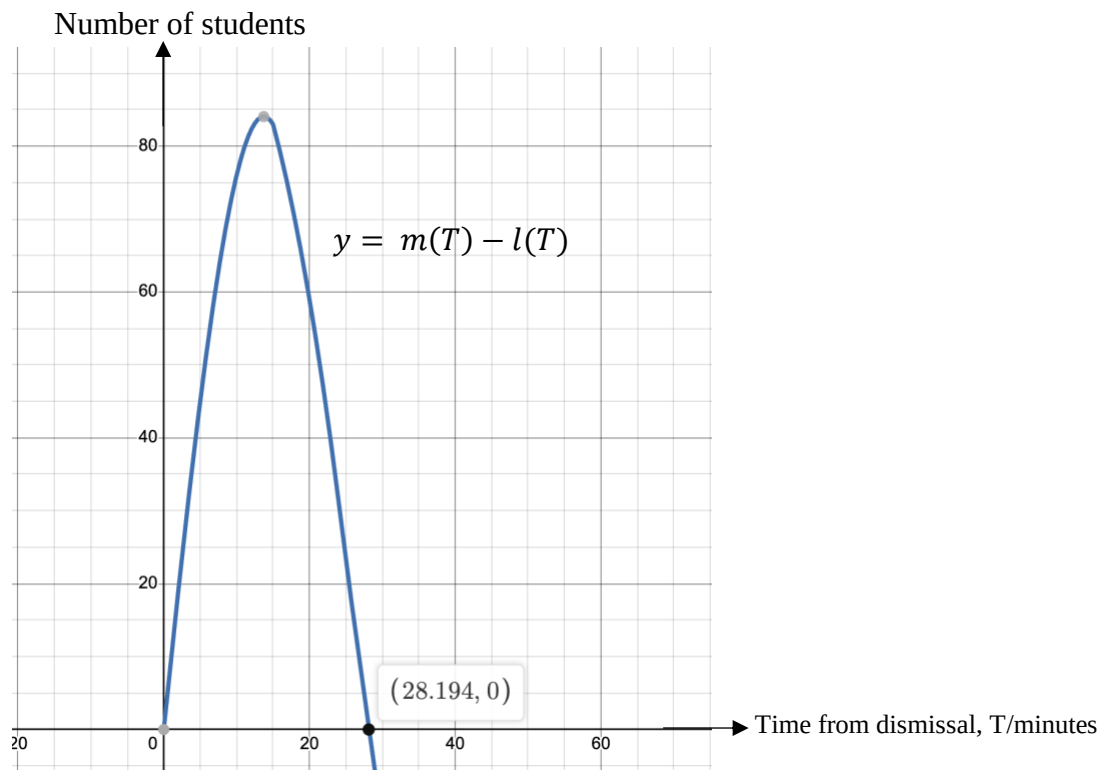


Figure 10. Area under the graph of number of students at the bus stop against time T provides total student time wasted.

Number of students at bus stop with respect to time from dismissal is,

$$\sum_{r=1}^{\frac{T}{2.5}} f(r) - \sum_{r=1}^{\frac{T}{5}} g(r) = m(T) - l(T)$$

Total student time wasted is,

$$\int_0^T m(T) - l(T)$$

For $0 \leq T \leq 15$, total student time wasted is,

$$\begin{aligned} & \int_0^T \sum_{r=1}^{\frac{T}{2.5}} f(r) - \sum_{r=1}^{\frac{T}{5}} g(r) \\ &= \int_0^T \frac{-8T^3 + 7550T}{750} - \frac{2T^3 + 60T^2 + 250T}{750} \end{aligned}$$

$$\begin{aligned}
&= \int_1^T \frac{-10T^3 - 60T^2 + 7300T}{750} \\
&= -\frac{1}{75} \int_1^T T^3 - \frac{2}{25} \int_1^T T^2 + \frac{146}{15} \int_1^T T \\
&= \left[-\frac{1}{75} \left(\frac{T^4}{4} \right) - \frac{2}{25} \left(\frac{T^3}{3} \right) + \frac{146}{15} \left(\frac{T^2}{2} \right) \right]_1^T \\
&= \left[-\frac{T^4}{300} - \frac{2T^3}{75} + \frac{146T^2}{30} \right]_1^T
\end{aligned}$$

For example, when $T = 10$, total time wasted is,

$$\begin{aligned}
&\left[-\frac{T^4}{300} - \frac{2T^3}{75} + \frac{146T^2}{30} \right]_1^{10} \\
&= \frac{(10)^4}{300} - \frac{2(10)^3}{75} + \frac{146(10)^2}{30} - \left(-\frac{1}{300} - \frac{2}{75} + \frac{146}{30} \right) \\
&= 488 \text{ minutes (to 3sf)}
\end{aligned}$$

For $15 < T \leq 25$, total student time wasted is

$$\begin{aligned}
&\int_1^T \sum_{r=1}^{\frac{T}{2.5}} f(r) - \sum_{r=1}^{\frac{T}{5}} g(r) \\
&= \int_1^T 105 + \frac{4T}{5} - \frac{2T^3 + 60T^2 + 250T}{750} \\
&= -\frac{2}{750} \int_1^T T^3 - \frac{60}{750} \int_1^T T^2 + \frac{53}{35} \int_1^T T + \int_1^T 105 \\
&= \left[-\frac{1}{375} \left(\frac{T^4}{4} \right) - \frac{2}{25} \left(\frac{T^3}{3} \right) + \frac{53}{35} \left(\frac{T^2}{2} \right) + 105T \right]_1^T \\
&= \left[-\frac{T^4}{1500} - \frac{2T^3}{75} + \frac{53T^2}{35} + 105T \right]_1^T
\end{aligned}$$

For example, when $T = 20$ total time wasted is,

$$\begin{aligned}
&\left[-\frac{T^4}{1500} - \frac{2T^3}{75} + \frac{53T^2}{35} + 105T \right]_1^{20} \\
&= \frac{(20)^4}{1500} - \frac{2(20)^3}{75} + \frac{53(20)^2}{35} + 105(20) - \left(-\frac{1}{1500} - \frac{2}{75} + \frac{53}{35} + 105 \right) \\
&= 542 \text{ minutes (to 3sf)}
\end{aligned}$$

Question 3. What is the probability of a student being able to board a bus immediately at time T?

Number of students waiting at bus stop at time T is, $m(T) - l(T)$

Number of buses required for boarding immediately, k, is given by,

$$[m(T) - l(T)] = k \left(g \left(\frac{T}{5} \right) \right),$$

as the number of students at the bus stop is equal to k multiplied by the boarding capacity.

Hence,

$$k = \frac{[m(T) - l(T)]}{\left(g \left(\frac{T}{5} \right) \right)}$$

Number of buses
required for boarding
immediately / k

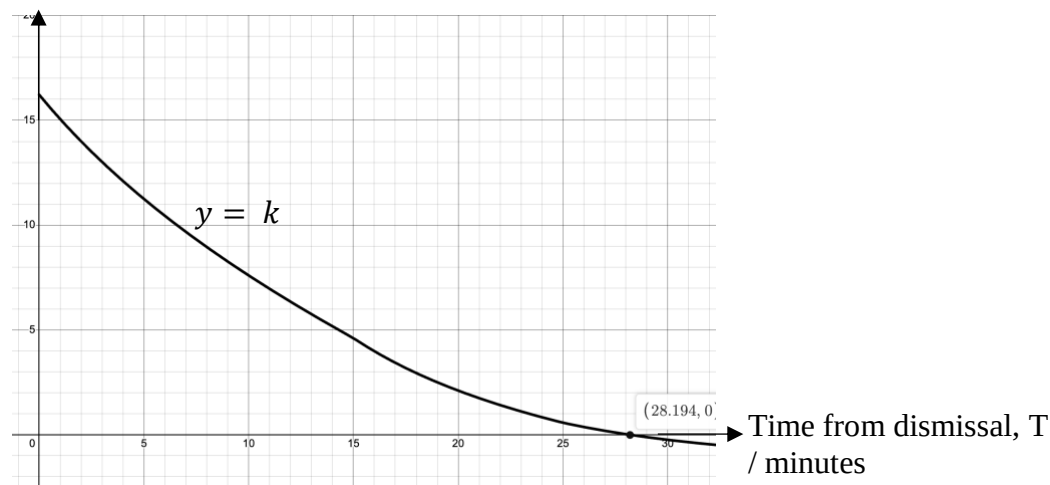


Figure 11. Number of buses needed for all students to board, with respect to time from dismissal

Following the Poisson distribution used earlier in part 1 of the exploration, we can substitute values of k based on the graph to find probability of students being able to board a bus immediately at time T.

For example, from the graph in Figure 11, at T = 15, $k = 4.611 \cong 5$.

When k = 5,

$$\begin{aligned} \text{Probability of } k \text{ buses arriving, } P(k) &= e^{-2} \left(\frac{2^5}{5!} \right) \text{ where, } k \in \mathbb{R}, \quad \lambda \in \mathbb{Z}^+ \\ &= 3.61\% \text{ (to 3sf)} \end{aligned}$$

This small probability of boarding the bus, corresponds to the findings from Question 1. Question 1 found that students have a small chance of being able to board the bus immediately for the first 20 minutes from dismissal, as the number of students exceeds the boarding capacity of the bus. The findings also correspond to observations, as I have never been able to board a bus immediately, within 20 minutes from dismissal.

Limitations

Firstly, the model assumes that the rate at which students arrive at the bus stop and board the bus will follow the modelled functions. This is based on a very limited sample size of 30 data points over 6 days, which may lead to results deviating from the real world scenarios of bus arrival and boarding. For example, the traffic conditions and bus arrival may differ depending on the day of the week, or even the week itself. Additionally, in situations where students are dismissed earlier than usual, this model will break down, as it assumes a constant time of dismissal for all students. Since the boarding capacity of the bus, and thus the rate of student boarding, is very much dependant on the time of day, this assumption is critical for this model.

To improve the accuracy and precision of the modelled functions, data collection across multiple days can be conducted. This will aid in improving the functions to model the bus arrival, and student boarding.

Secondly, this model assumes that the rate at which buses arrive is constant. Although the derived rate of one bus every five minutes is quite accurate, having been validated using Poisson distribution, the effect of traffic or other factors that affect the rate of bus arrival is not accounted for. This will again leading to a deviation from real world scenarios. For example, in cases of traffic due to poor weather, buses may arrive less frequently.

To mitigate such limitations, the investigation can be extended by finding how each parameter affects bus arrival using sensitivity analysis. Additionally, to account for the element of random variation in terms of bus arrival, Monte Carlo simulations⁷ can be used to find the probability of different rates of bus arrival. By introducing random variation in the modelled parameters such as weather, congestion, vehicle breakdowns, the simulation can be used to find different outcomes of bus arrival rate. Thus, running specialised computer simulations to find the probability of different rates of bus arrival more accurately determines what is the most likely rate of bus arrival.

⁷ Santos, G. (2022, May 24). *A Gentle Introduction to the Monte Carlo Simulation*. Medium.
<https://towardsdatascience.com/a-gentle-introduction-to-the-monte-carlo-simulation-a3e5518a4aa7>

Conclusion

In this exploration, I set out with the objective of modelling student arrival and boarding patterns at my school bus stop, to minimise idle student waiting time. I used the Poisson Distribution to validate the rate of bus arrival, as well as sequences and summation to model student arrival and boarding. The model provided concrete results that students should start waiting for the bus at 17.5 minutes from dismissal only, as well as information about total student time wasted.

These results are justified as this model takes into account the rate of bus arrival, the boarding patterns of students as well as the non-linear arrival of students and buses. Thus, this model provides valuable information to quantify the significant amount of time wasted at bus stops, and inform students of when to wait for the bus. For example, the model shows that after a mere 10 minutes from dismissal, total student time wasted is 488 minutes. Thus, using models such as this to optimise the use of student's time is extremely valuable. It also provides insights into more effective means of transport, or dismissal time for students. For example, schools may consider staggered dismissal times for different levels or different classes, to ensure that students can at least spend their time productively in school, rather than waiting idly at the bus stops.

Moreover, with regards to the times when students should wait for the bus, the rather counter intuitive result that for the first 17.5 minutes, students are better off walking to the train, vindicates the value in such models. Intuition is not fool-proof, and it can lead to unintended consequences, such as wasted time, and false assumptions. Thus, where intuition may fail us in making good decisions, mathematical modelling provides more accurate information for us to make informed choices.

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Appendix

Day 1

1 st interval	2 (buses)
2 nd interval	3
3 rd interval	2
4 th interval	1
5 th interval	4

Day 2

1 st interval	2
2 nd interval	1
3 rd interval	1
4 th interval	3
5 th interval	2

Day 3

1 st interval	2
2 nd interval	2
3 rd interval	2
4 th interval	3
5 th interval	1

Day 4

1 st interval	3
2 nd interval	2

3 rd interval	2
4 th interval	3
5 th interval	4

Day 5

1 st interval	4
2 nd interval	1
3 rd interval	3
4 th interval	1
5 th interval	2

Day 6

1 st interval	3
2 nd interval	4
3 rd interval	3
4 th interval	2
5 th interval	0